Helical Gear Design – Analytical Calculations Report

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Analytical methods (Calculation)

1. Gear Design Parameters

Gear type: Helical gears (parallel shafts)
Normal module (mn) = 2.5 mm
Helix angle (β) = 20°
Normal pressure angle (φn) = 20°
Number of teeth: z1 = 24 (pinion), z2 = 48 (gear)
Face width (b) = 30 mm
Input torque (T) = 20 N·m
Pinion speed = 1500 rpm

2. Gear Geometry Calculations

Transverse module:
$mt = mn / \cos(\beta)$
$mt = 2.5 / \cos(20°) = 2.66$ mm

Pitch diameters:
$d1 = mt \times z1 = 2.66 \times 24 = 63.85$ mm
$d2 = mt \times z2 = 2.66 \times 48 = 127.70$ mm

Center distance:
$a = (d1 + d2) / 2$
$a = (63.85 + 127.70) / 2 = 95.78$ mm

3. Tangential Force on Gear Teeth

Formula:
$Ft = 2T / d1$

Convert torque:
T = 20 N·m = 20000 N·mm

$Ft = (2 \times 20000) / 63.85$
$Ft = 626.5$ N

4. Helical Gear Forces

Axial force:
$Fa = Ft \times \tan(\beta)$
$Fa = 626.5 \times \tan(20°) = 228$ N

Transverse pressure angle:
$\tan(\varphi t) = \tan(\varphi n) / \cos(\beta)$
$\varphi t \approx 21.17°$

Radial force:
$Fr = Ft \times \tan(\varphi t)$
$Fr = 626.5 \times \tan(21.17°) = 242$ N

5. Bending Stress Calculation (Lewis Equation)

For helical gears, virtual number of teeth is used:
$zv = z / \cos^3(\beta)$
$zv = 24 / \cos^3(20°) = 28.92$

Lewis form factor:
$y = 0.154 - (0.912 / zv)$
$y = 0.1225$

Bending stress equation:
$\sigma b = Ft / (b \times mn \times y)$

$\sigma b = 626.5 / (30 \times 2.5 \times 0.1225)$
$\sigma b = 68.2$ MPa

6. Including Dynamic and Load Factors

Factors used:
$Ko = 1.0$
$Km = 1.2$
Dynamic Factor (AGMA) Kv Calculation

$Qv = 10$
1) Pitch-line velocity: $V\ (m/s) = (\pi\ d1\ n) / 60 = (\pi \times 0.06385 \times 1500) / 60 = 5.015$ m/s
Convert to ft/min: $V\ (ft/min) = 5.015 \times 196.85 = 987$ ft/min
2) AGMA constants:
$B = 0.25\ (12 - Qv)^{(2/3)} = 0.25\ (12 - 10)^{(2/3)} = 0.25\ (2)^{(2/3)} = 0.3969$
$A = 50 + 56(1 - B) = 50 + 56(1 - 0.3969) = 50 + 56(0.6031) = 83.77$
3) Dynamic factor:
$Kv = [(A + \sqrt{V}) / A]^B$
$= [(83.77 + \sqrt{987}) / 83.77]^{0.3969}$
$= [(83.77 + 31.42) / 83.77]^{0.3969} = (1.375)^{0.3969} = 1.135$
Result : $Kv \approx 1.135$

Effective bending stress:
$\sigma b_eff = \sigma b \times (Ko \times Kv \times Km)$
$\sigma b_eff = 68.2 \times (1 \times 1.135 \times 1.2)$
$\sigma b_eff = 92.9$ MPa

7. Contact Stress (Hertz Theory)

Contact stress formula:
$\sigma c = Cp \sqrt{[(Kv \times Wt / (b \cos\varphi))\ (1/r1 + 1/r2)]}$

Elastic coefficient:
$Cp = sqrt(1 / [\pi((1-\nu1^2)/E1 + (1-\nu2^2)/E2)])$

For steel–steel:
$Cp \approx 191.65$

Using calculated parameters:
$\sigma c \approx 348.5$ MPa

8. Material Selection

Material option used for evaluation:
AISI 4140 / 42CrMo4 (quenched and tempered)
Hardness ≈ 300 HB

Allowable bending stress ≈ 300 MPa
Surface endurance limit:
$Sc = 2.8 \times HB$
$Sc = 2.8 \times 300 = 840$ MPa

9. Safety Factor Calculations

Bending safety factor:
$SF_b = \sigma_allow / \sigma b_eff$
$SF_b = 300 / 92.9$
$SF_b \approx 3.23$

Contact fatigue safety factor:
$SF_c = (Sc / \sigma c)^2$
$SF_c = (840 / 348.5)^2$
$SF_c \approx 5.8$

10. Summary of Results

Tangential force Ft = 626.5 N
Axial force Fa = 228 N
Radial force Fr = 242 N
Bending stress = 92.9 MPa
Contact stress = 348.5 MPa
Bending safety factor ≈ 3.2
Contact safety factor ≈ 5.8

2D Finite Element Analysis Attempt

An initial finite element analysis was performed using a two-dimensional (2D) model in order to reduce the computational cost and overcome the hardware limitations encountered during the full 3D simulation.

However, the obtained results were found to be significantly different from the expected analytical values. This discrepancy is mainly due to the fact that the analyzed gear pair is a helical gear system, whose behavior is inherently three-dimensional.

In helical gears, the load acting on the tooth is resolved into three components:

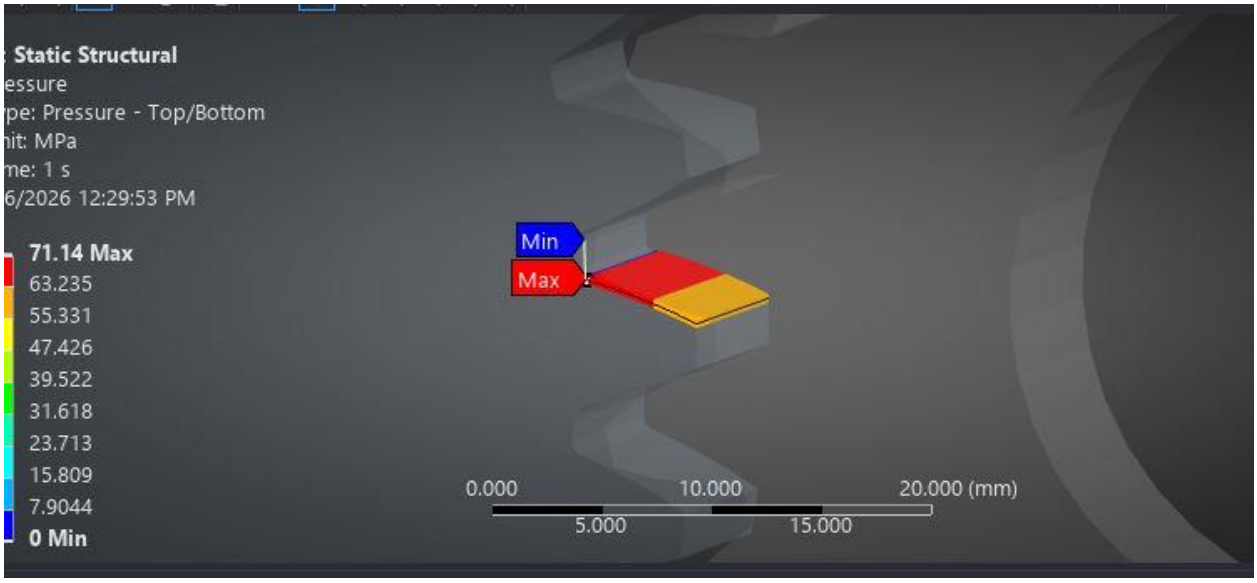
- Tangential force Wt
- Radial force Wr
- Axial force Wa

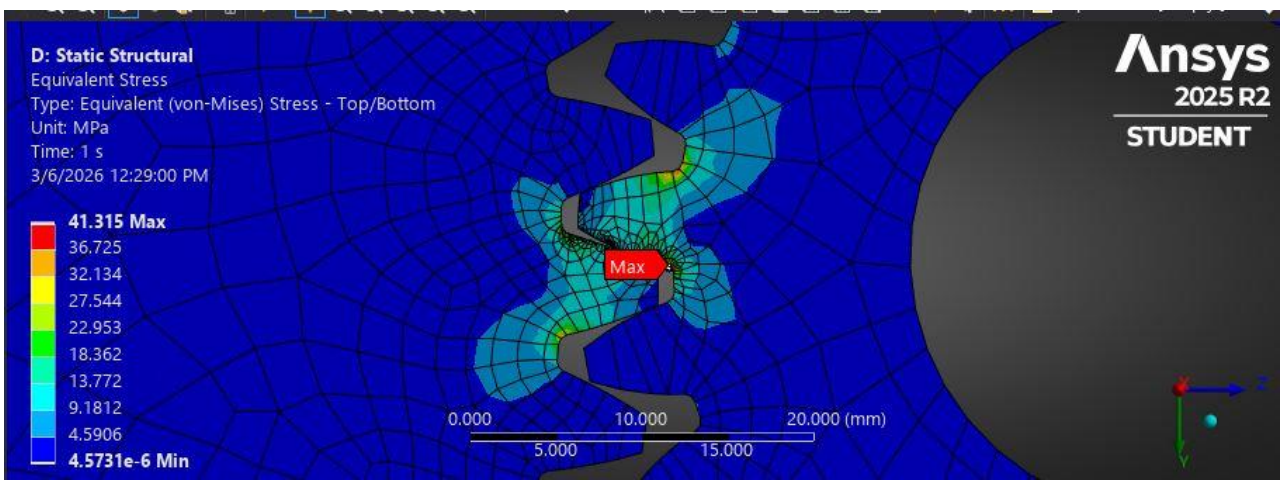
The axial component, which arises from the helix angle, plays an important role in the load distribution along the face width and significantly influences both the bending stress and the contact pressure distribution.

- In a 2D analysis, the geometry is reduced to a planar representation, which means that:
- The helix angle cannot be represented.
- The axial load component is completely neglected.
- The load distribution along the face width is not captured.
- The tooth contact is simplified into a line contact instead of a three-dimensional contact patch.

Because of these limitations, the 2D model effectively behaves like an equivalent spur gear model, which does not accurately represent the true mechanical behavior of a helical gear pair. As a result, the predicted stresses and contact pressures were not reliable for validating the analytical calculations.

Therefore, a three-dimensional model was later adopted to better represent the actual geometry and loading conditions of the helical gear system.





3D Finite Element Model Development

After observing the limitations of the 2D analysis for representing the behavior of a helical gear system, a three-dimensional finite element model was developed in ANSYS to better capture the real contact conditions between the gear teeth.

1. Geometry Simplification and Sector Extraction

Since performing a full 3D simulation of the entire gear pair requires significant computational resources, a reduced sector model was created to represent the contact behavior while maintaining reasonable computational cost.

The following steps were performed on the geometry:

- A sector containing three teeth from each gear was extracted.
- The cut was performed along radial planes passing through the gear axis, preserving the tooth of interest and its adjacent teeth.
- A portion of the gear body behind the teeth was retained to maintain realistic stiffness at the tooth root.

This approach allows the model to preserve the local mechanical behavior of the gear tooth, while significantly reducing the number of elements required for the simulation.

2. Contact Definition Between Gear Teeth

To simulate the interaction between the gear teeth, a contact pair was defined between the mating tooth flanks.

The contact setup included:

- Contact type: Frictionl contact
- Fraction Factor : 0.15
- Contact formulation: Augmented Lagrange
- Interface treatment: Adjust to Touch

3. Mesh Refinement

A refined mesh was applied in critical regions to accurately capture the stress distribution. The mesh strategy included:

- A coarser mesh in the bulk regions of the gears.
- Local mesh refinement at the tooth contact surfaces to accurately capture the contact pressure distribution.
- Additional refinement at the tooth root fillet, where the bending stress is expected to reach its maximum value.

This local refinement ensures an accurate representation of both:

- Contact stresses at the tooth surface

- Bending stresses at the tooth root

4. Boundary Conditions and Loading

To represent the operating conditions of the gear pair, appropriate boundary conditions were applied.

- **Gear Support**
  The driven gear was constrained using a fixed support applied at the bore surface, representing the shaft constraint.
- **Pinion Loading**
  The driving gear (pinion) was subjected to a moment applied at the bore, representing the transmitted torque.

The applied torque corresponds to the same loading conditions used in the analytical calculations.

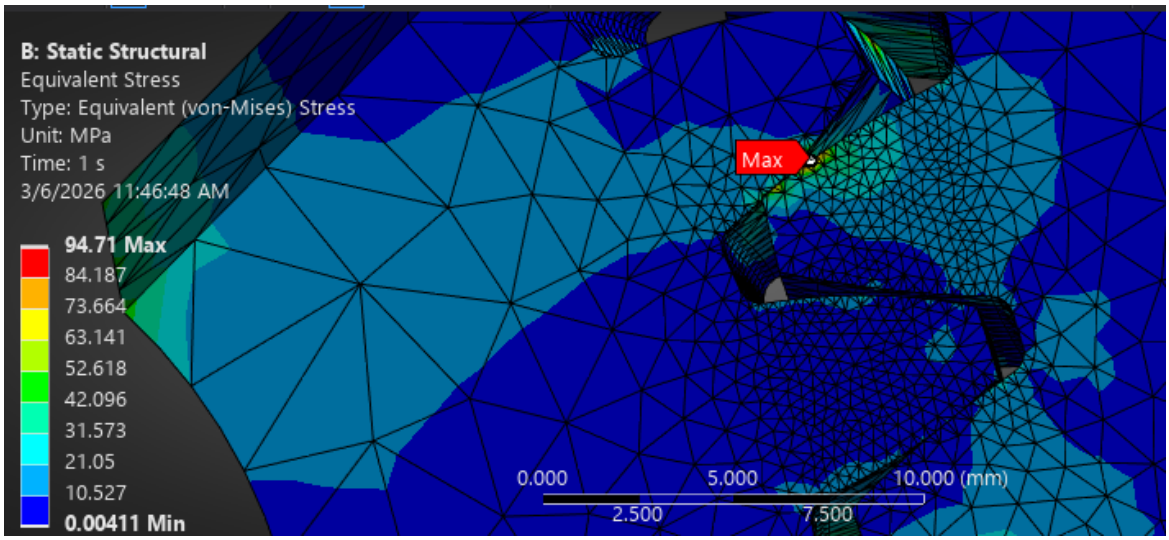
5. Finite Element Results

The simulation results show that the maximum equivalent (von Mises) stress occurs near the tooth root, which is consistent with the expected bending stress location in gear teeth. The obtained maximum stress value was approximately:

σ max ≈94.7MPa

This value is very close to the analytical bending stress calculated earlier, confirming the validity of both the analytical approach and the finite element model.

The stress concentration observed at the tooth root also agrees with classical gear design theory, where the tooth root acts as the critical section for bending failure.



6. Pitting Surface Stress (Surface Contact Stress)

The contact interaction between the mating tooth flanks was analyzed using the Contact Tool in ANSYS, where the maximum contact pressure at the tooth interface was extracted.

The simulation results indicated that the maximum contact pressure occurred within the contact region between the mating tooth surfaces, forming a localized contact patch typical for gear tooth engagement.
The obtained maximum contact pressure was approximately:

σc ≈360 MPa

This value is very close to the analytically calculated pitting surface stress, confirming the consistency between the finite element model and the theoretical calculations.

The results indicate that the surface stresses remain within the allowable range for the selected material, suggesting that the gear teeth are not expected to fail due to surface pitting under the applied loading conditions.

7. Discussion

The close agreement between the analytical calculations and the finite element results indicates that the developed model successfully captures the main mechanical behavior of the gear teeth under load.
The 3D model also provides a clearer visualization of the stress distribution within the tooth structure, especially in regions where analytical methods provide only approximate estimates.

Conclusion

In this study, the mechanical behavior of a **helical gear pair** was investigated using both **analytical calculations and finite element analysis (FEA)**.

Initially, the bending stress and surface contact stress were estimated using classical gear design equations. These calculations provided preliminary values for the **tooth bending**

stress and the **pitting surface stress**, which were used as reference values for validating the numerical model.

A finite element model was then developed in **ANSYS** to evaluate the stress distribution within the gear teeth. Due to computational limitations, an initial **2D model** was considered; however, this approach was found to be insufficient for representing the behavior of **helical gears**, since the helix angle and the associated axial load component cannot be captured in a planar model.

Therefore, a **3D sector model containing several teeth** was created to better represent the actual gear geometry and tooth contact conditions while maintaining a manageable computational cost.

The finite element results showed that the **maximum von Mises stress occurs at the tooth root**, which agrees with classical gear theory where the tooth root acts as the critical section for bending failure. The obtained stress values were found to be **close to the analytical predictions**, confirming the validity of the developed model.

In addition, the **contact pressure distribution between the mating tooth flanks** was evaluated to assess the risk of surface pitting. The calculated contact stress and the corresponding safety factor indicated that the gear teeth are capable of operating safely under the applied loading conditions.

Overall, the comparison between **analytical calculations and numerical simulation** demonstrates that the adopted design approach provides a reliable estimation of both bending and surface stresses in the gear teeth.